Peyman Jabbarzade

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Education

Ph.D. in Computer Science, University of Maryland – College Park, US — Jan 2022 – present

- **Field:** Theoretical Computer Science
- Designing algorithms for fundamental graph problems and optimizing submodular functions in dynamic settings.

M.Sc. in Computer Engineering, Sharif University of Technology – Tehran, Iran — Sept 2019 – Dec 2021

- **Field:** Algorithms and Computation
- Studied geometric graph problems in the massively parallel computation (MPC) model.

B.Sc. in Computer Engineering, Sharif University of Technology – Tehran, Iran — Sept 2015 – Aug 2019

Awards

Dean's Fellowship, University of Maryland — Jan 2022 – Dec 2023

Asia West Champion, International Collegiate Programming Contest (**ICPC**) World Final — Apr 2018

14th Place, ACM International Collegiate Programming Contest (**ACM-ICPC**) World Final — May 2016

Bronze Medal, International Olympiad in Informatics (**IOI**) — July 2015

Gold Medal, Asia-Pacific Informatics Olympiad (**APIO**) — May 2015

Gold Medal, Iranian National Olympiad in Informatics — Sept 2014

Experience

Software Engineer, Balad (Map and Navigation App) – Tehran, Iran — Sept 2018 – Nov 2021

- Enhanced service stability and scalability and optimized inter-service communication as part of the infrastructure team.
- Developed and took ownership of Balad's core service for managing and providing location-specific data.
- Led a diverse team of 10 engineers to collect and clean location-specific data, and to refine the service design using microservices.
- Primarily coded in Python, with additional experience in C++, Java, Go, and various other tools.

Research Intern, Max Planck Institute for Informatics – Saarbrücken, Germany — July 2019 – Aug 2019

- Designed and benchmarked an energy-preserving online scheduling algorithm.

Software Engineer Intern, Balad (Map and Navigation App) – Tehran, Iran — June 2018 – Aug 2018

- Designed and implemented a novel algorithm for optimal routing of camera-equipped cars, ensuring comprehensive street coverage in Tehran within the minimum time.

Publications

(All papers use **alphabetical** author ordering.)

Breaking a Long-Standing Barrier: 2-ε Approximation for Steiner Forest — Apr 2025
Ali Ahmadi, Iman Gholami, MohammadTaghi Hajiaghayi, Peyman Jabbarzade, Mohammad Mahdavi

Dynamic Algorithms for Submodular Matching — Apr 2025
Kiarash Banihashem, Leyla Biabani, Samira Goudarzi, MohammadTaghi Hajiaghayi, Peyman Jabbarzade, Morteza Monemizadeh
52nd EATCS International Colloquium on Automata, Languages, and Programming (**ICALP 2025**).

Prize-Collecting Forest with Submodular Penalties: Improved Approximation — Jan 2025
Ali Ahmadi, Iman Gholami, MohammadTaghi Hajiaghayi, Peyman Jabbarzade, Mohammad Mahdavi
26th Conference on Integer Programming and Combinatorial Optimization (**IPCO 2025**).

A Dynamic Algorithm for Weighted Submodular Cover Problem — July 2024
Kiarash Banihashem, Samira Goudarzi, MohammadTaghi Hajiaghayi, Peyman Jabbarzade, Morteza Monemizadeh
41st International Conference on Machine Learning (**ICML 2024, accepted for oral presentation)**

Prize-Collecting Steiner Tree: A 1.79 Approximation — June 2024
Ali Ahmadi, Iman Gholami, MohammadTaghi Hajiaghayi, Peyman Jabbarzade, Mohammad Mahdavi
56th ACM Symposium on Theory of Computing (**STOC 2024**).

2-Approximation for Prize-Collecting Steiner Forest Jan 2024
Ali Ahmadi, Iman Gholami, MohammadTaghi Hajiaghayi, Peyman Jabbarzade, Mohammad Mahdavi
Journal of the ACM (**JACM 2025**). ACM-SIAM Symposium on Discrete Algorithms (**SODA 2024, selected for special issue**).

Dynamic Algorithms for Matroid Submodular Maximization Jan 2024
Kiarash Banihashem, Leyla Biabani, Samira Goudarzi, MohammadTaghi Hajiaghayi, Peyman Jabbarzade, Morteza Monemizadeh
ACM-SIAM Symposium on Discrete Algorithms (**SODA 2024**).

Dynamic Non-monotone Submodular Maximization Dec 2023
Kiarash Banihashem, Leyla Biabani, Samira Goudarzi, MohammadTaghi Hajiaghayi, Peyman Jabbarzade, Morteza Monemizadeh
37th Conference on Neural Information Processing Systems (**NeurIPS 2023**).

Dynamic Constrained Submodular Optimization with Polylogarithmic Update Time July 2023
Kiarash Banihashem, Leyla Biabani, Samira Goudarzi, MohammadTaghi Hajiaghayi, Peyman Jabbarzade, Morteza Monemizadeh
40th International Conference on Machine Learning (**ICML 2023**)

A Novel Prediction Setup for Online Speed-Scaling June 2022
Antonios Antoniadis, Peyman Jabbarzade, Golnoosh Shahkarami
18th Scandinavian Symposium and Workshops on Algorithm Theory (**SWAT 2022**)

Volunteer Experience

Paper Reviewing 2020 – present

- Served on the Junior Program Committee for SPAA 2025.
- Reviewed papers for Algorithmica journal and conferences such as DISC 2022, ALGOSENSORS 2022, NeurIPS 2024, IPCO 2024, APPROX 2024, ESA 2024, ICML 2025, ICLR 2025, AISTATS 2025, ICALP 2025, and SPAA 2025.

Coaching 2020 – 2023

- Coached the University of Maryland's team for ICPC 2023, leading them to a silver medal at the North America Championship and advancing to the world final. [1, 2]
- Led Iran's national team for IOI 2020, achieving 3 gold and 1 silver medal—the best result for Iran in IOI.
- Coached Sharif University of Technology's team in the ICPC 2020 World Finals.

Scientific Committee 2017 – 2024

- Member of the Host Scientific Committee (HSC) for IOI 2017.
- Judge for the ICPC North America Championship 2024.
- Member of the Iranian National Olympiads in Informatics Scientific Committee over six years.
- Contributed to the preparation of the ICPC Tehran Regional Contest over six years.

Teaching 2015 – 2024

- Teaching Assistant at the University of Maryland for courses in Algorithms, Advanced Data Structures, and Computational Genomics.
- Teaching Assistant at Sharif University of Technology for courses in Design Algorithms and Discrete Structures.
- Prepared high school students for the Iranian National Olympiad in Informatics.

Task Author 2015 – 2024

- Authored the problem "Rectangles" for IOI 2019.
- Managed numerous programming competitions and authored problems for selecting Iran's national team for IOI.

Technical Skills

Programming Languages: Python, C/C++, SQL

Technologies: Docker, Kubernetes, Kafka, gRPC, REST, Sentry, Redis, Prometheus, Grafana, ElasticSearch, Logstash, Kibana